F12 4/4

Lecture 9: Section 1.5
Analyzing Graphs of Functions

Vertical line test for functions

Zeros of a function

Increasing and Decreasing Functions

Relative Minimum and maximum

Average rate of change

Even and odd functions

F12 4th

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$y = 2x + 3$

$f(x) = 2x + 3$

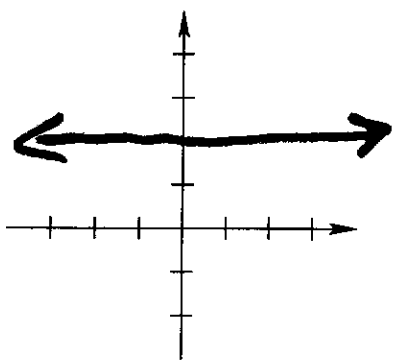
Lecture 9: Section 1.5
Analyzing Graphs of Functions

The **graph** of a function $y = f(x)$ is the set of points (x, y) in the xy-plane that satisfy the equation.

For each x, there is only one y.

<u>ex.</u> Which of the following graphs represent y as a function of x? Find the domain and range of each function.

1) $y = 2$



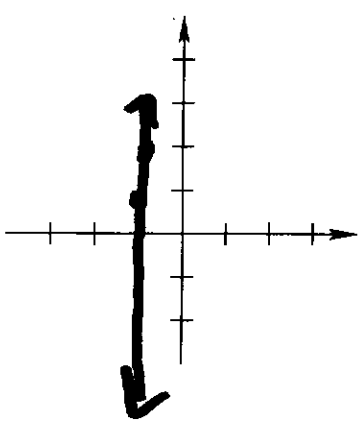
x	y
-1	2
0	2
1	2

Yes a function

D: all reals, $(-\infty, \infty)$

R: $\{2\}$

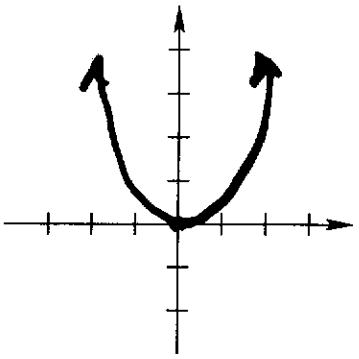
2) $x = -1$



x	y
-1	2
-1	1
-1	0

Not a function

3) $y = x^2$



x	y
2	4
1	1
0	0
-1	1
-2	4

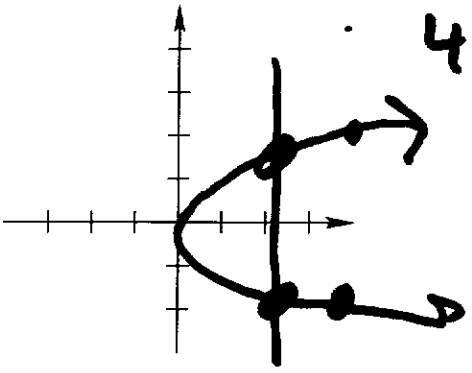
Function.

D: $(-\infty, \infty)$

R: $[0, \infty)$

y values

4) $x = y^2$



x	y
4	2
4	-2

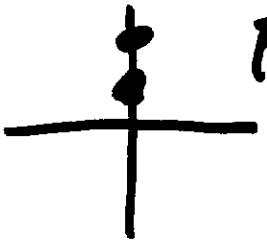
Not a function.

Vertical Line Test for Functions

A set of points in the xy-plane is the graph of y as a function of x if and only if **no vertical line** intersects the graph at more than one point.

Function if f any vertical line intersects the graph at most once.

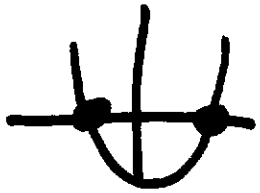
NOTE: A function can have at most one y-intercept.



Fails vert. line test.

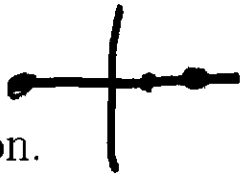
1

Zeros are where $y=0$, so on a graph the zeros are the x-intercepts.



Zeros of a Function

Def. The **zeros of a function** $f(x)$ are the x-values for which $f(x)=0$.



ex. Find the zeros of each function.

Let $y=0$

1) $f(x) = x^3 - x^2 - 4x + 4 = 0$

$(x^3 - x^2) - (4x - 4) = 0$

$x^2(x-1) - 4(x-1) = 0$

$(x-1)(x^2-4) = 0$

$x-1=0$

$x=1$

$x^2-4=0$

$x^2=4$

$x=\pm 2$

zeros are $\boxed{x=1, \pm 2}$

2) $g(x) = \left(\sqrt{4-x^2}\right)^2 = (0)^2$

$4-x^2=0$

$4=x^2$

$x=\pm 2$

Must check for extr. sol.

$x=2$ ✓

$x=-2$ ✓

3) $h(x) = \dfrac{1-2x}{1+3x} = 0$

$\dfrac{1-2x}{1+3x} = \dfrac{0}{1}$

$0 = 1-2x$

$1-2x=0$

$-2x=-1$

$\boxed{x=\frac{1}{2}}$ Make sure sol. is in the D.

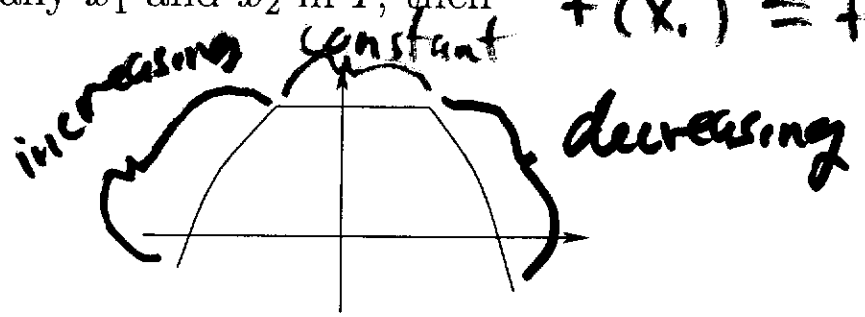
$\dfrac{\quad}{7} = 0$

Increasing and Decreasing Functions

Def. A function f is **increasing** on an open interval I if for any x_1 and x_2 in I, if $x_1 < x_2$, then $f(x_1) < f(x_2)$

Def. A function f is **decreasing** on an open interval I if for any x_1 and x_2 in I, if $x_1 < x_2$, then $f(x_1) > f(x_2)$

Def. A function f is **constant** on an open interval I if for any x_1 and x_2 in I, then $f(x_1) = f(x_2)$



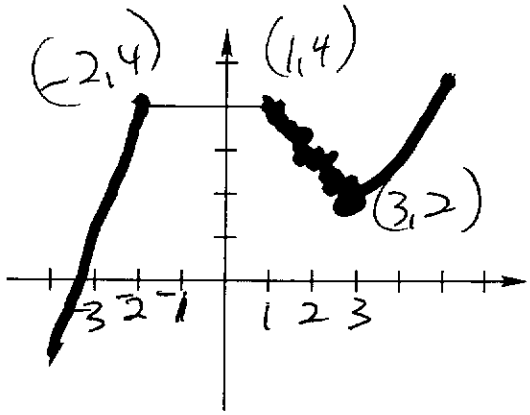
Relative Minimum and Maximum

Def. A function has a **relative (local) minimum** at $(a,\ f(a))$, if there is an open interval I containing a so that, for all $x \neq a$ in I, $f(x) \geq f(a)$

Def. A function has a **relative (local) maximum** at $(a,\ f(a))$, if there is an open interval I containing a so that, for all $x \neq a$ in I, $f(x) \leq f(a)$

highest pt on the graph in that vicinity.

ex. Consider the following graph $y = f(x)$ and find the following:



y-values

1) Interval(s) on which f is increasing $(-\infty, -2) \cup (3, \infty)$

2) Interval(s) on which f is decreasing $(1, 3) \rightarrow 1 < x < 3$

3) Interval(s) on which f is constant (flat) $(-2, 1) \rightarrow -2 < x < 1$

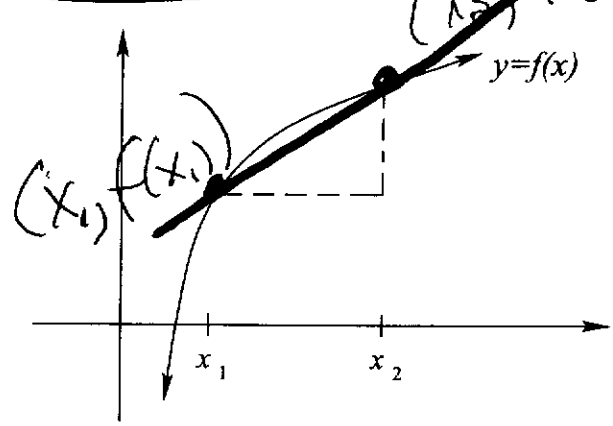
4) Relative minimum value(s) $(3, 2)$ or $f(3) = 2$

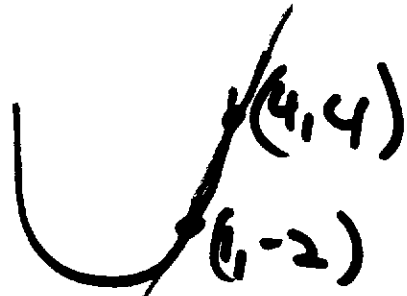
5) Relative maximum value is 4. It occurs between $x = -2$ and $x = 1$.

Average Rate of Change

<u>**Def.**</u> **The average rate of change** of a function from x_1 to x_2 is the slope of the **secant line** through the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$.

$$m = \frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}, x_1 \neq x_2$$





slope of secant line

(1,-2)
(4,4)

ex. Given $f(x) = x^2 - 3x$.

1) Find the average rate of change of $f(x)$ from 1 to 4.

$$A.R.C = \frac{f(4)-f(1)}{4-1} = \frac{4-{-2}}{4-1} = \frac{6}{3} = \boxed{2 = \frac{2}{1}}$$

2) Write the equation of the secant line through the points $(1, f(1))$ and $(4, f(4))$.

1. Find slope of secant line.
   From above $m = 2$
2. Plug in one of the coordinates.

$$y - y_1 = m(x - x_1)$$

$$y - 4 = 2(x-4)$$

$$\boxed{y = 2x - 4}$$

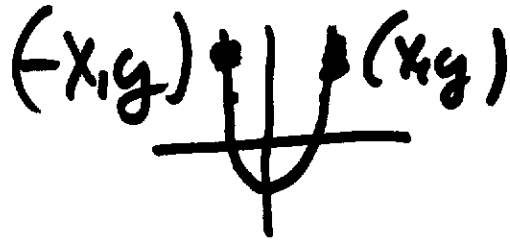
(4,4)

3) Find the average rate of change of $f(x)$ from 1 to x.

$$A.R.C. = \frac{f(x)-f(1)}{x-1} = \frac{x^2-3x-(-2)}{x-1} = \frac{x^2-3x+2}{x-1}$$

$$= \frac{(x-2)(x-1)}{x-1} = \boxed{x-2}$$

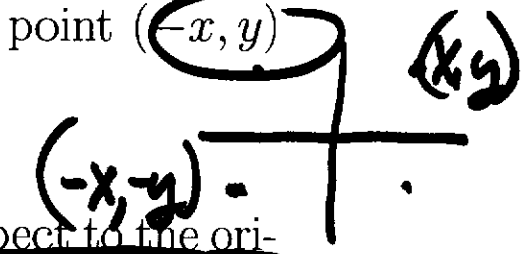
$f(1) = 1^2 - 3(1) =$
$= 1 - 3$
$= -2$



Even and Odd Functions

Def. If a graph is symmetric with respect to the y-axis, then the function is **even** and $f(-x) = f(x)$ for each x in the domain.

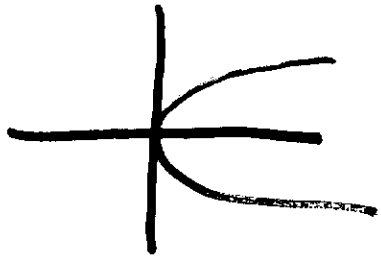
NOTE: $f(x)$ is even if and only if, whenever the point (x, y) is on the graph of f, the point $(-x, y)$ is also on the graph.



Def. If a graph is symmetric with respect to the origin, then the function is **odd** and $f(-x) = -f(x)$ for each x in the domain.

NOTE: $f(x)$ is odd if and only if, whenever the point (x, y) is on the graph of f, the point $(-x, -y)$ is also on the graph.

NOTE: What about symmetry with respect to the x-axis?



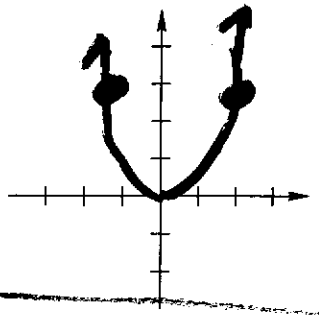
Not a function.

Even iff $f(-x) = f(x)$

odd iff $f(-x) = -f(x)$

ex. Determine if the function is even, odd, or neither. Graph the function.

1) $f(x) = x^2$ Even?



$f(-x) \stackrel{?}{=} f(x)$

$(-x)^2 =$

$x^2 = x^2$ ✓ so even

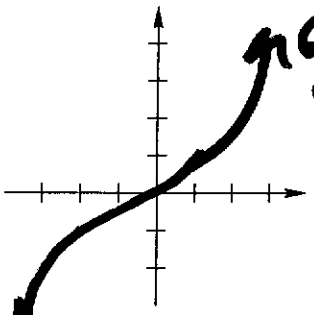
~~odd?~~

$f(-x) \stackrel{?}{\neq} -f(x)$

$(-x)^2 \neq -x^2$

$x^2 \neq -1 \cdot x^2$

2) $g(x) = x^3$ ~~Even?~~



$g(-x) \stackrel{?}{\neq} g(x)$

$(-x)^3 \neq$

$-x^3 \neq x^3$

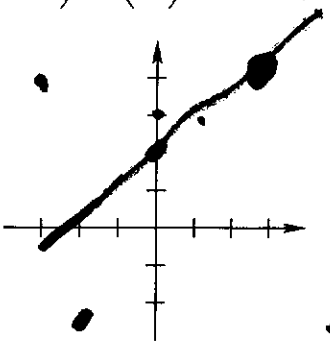
odd

$g(-x) \stackrel{?}{=} -g(x)$

$(-x)^3 = -x^3$

$-x^3 = -x^3$ ✓

3) $h(x) = x + 2$



~~Even~~

$h(-x) \stackrel{?}{\neq} h(x)$

$-x+2 \neq x+2$

not even

~~odd~~

$h(-x) \neq -h(x)$

$-x+2 \neq -(x+2)$

$-x+2 \neq -x-2$